

MSc Project Proposal



MSc Data Science and AI Project Proposal

Shiyas Rasi

M01087756

Programme of Study

Data Science

Name of Supervisor

Sudhakar Camilos Fernando

The All-Rounder

An End-to-End Data Engineering and Predictive Framework for Demand Forecasting and Inventory Optimization in Pharmaceutical Distribution

Submission Date

10- Jul- 2026

Table of Contents

Table of Contents

Table of Contents	2
1. Abstract.....	3
2. Introduction.....	3
3. State of the Art Review	4
4. Methods.....	5
5. Proposed Work	5
5.1 Intended contribution of the work.....	5
5.2 Benefits of the proposed work	5
5.3 Procedures and activities of the proposed work	6
5.4 Evaluation of the project.....	6
5.5 Resources needed.....	6
5.6 Access to participants and clients	6
5.7 Ethical Aspects.....	7
6 Work plan and schedule.....	7
7 Conclusions	8
8 Bibliography.....	9
9 Appendixes.....	9

1. Abstract

The research goal of this project is to optimize supply chain efficiencies within the wholesale pharmaceutical distribution sector through the development of an automated, end-to-end data engineering pipeline and predictive AI framework. Utilizing a robust 5-year historical dataset containing transactional order ledgers, this study builds scalable architectures to systematically clean, process, and ingest high-dimensional stock data. Advanced machine learning models—specifically gradient boosting algorithms (e.g., XGBoost) and classical time-series models—will be implemented to accurately forecast SKU-level demand variations. The outputs will feed directly into an inventory optimization layer to dynamically compute safety stock and automated reorder points (*ROP*). The ultimate objective is to mitigate the dual logistical constraints of stockouts and product expiration waste, providing an actionable operational roadmap for modern pharmaceutical logistics.

2. Introduction

Modern pharmaceutical supply chains are tightly regulated networks that face distinct logistical challenges, such as volatile demand fluctuations, strict temperature controls, and finite product shelf lives. Inefficient inventory management leads either to catastrophic stockouts of life-saving medicines or expensive capital traps caused by expired overstocks. Traditional forecasting approaches often rely on basic rolling averages or static heuristic methods that fail to adapt to complex seasonal disease spikes, geographic anomalies, and dynamic order patterns.

This study introduces a comprehensive data-driven solution that shifts wholesale pharmacy distribution from reactive stocking to proactive, AI-guided replenishment. By utilizing an extensive 5-year ledger of historical order data, this project addresses structural inefficiencies in matching stock levels to regional health demands through advanced data pipelines and optimized time-series predictions.

- **Aims and Objectives**

- Aim 1: Build a scalable, automated data engineering pipeline for processing large-scale transactional pharmaceutical data.**

- 1.1:** Clean, normalize, and schema-map 5 years of historical order ledgers to resolve missing entries, clerical errors, and inconsistencies.
 - 1.2:** Aggregate data structurally by timestamp, geographic location, client tier, and product category (aligning with therapeutic classifications).
 - 1.3:** Build an optimized ETL/ELT architecture to automate the transition of raw data into analytical features.
 - 1.4:** Conduct Exploratory Data Analysis (EDA) to map non-linear demand trends, outliers, and seasonal variances across product tiers.

Aim 2: Develop and validate predictive machine learning frameworks for multi-level demand forecasting and inventory replenishment.

2.1: Implement and evaluate classical time-series baselines (such as SARIMA) against advanced machine learning algorithms (such as XGBoost/CatBoost).

2.2: Engineer predictive features incorporating temporal factors (e.g., month, season indicators) and historical lags.

2.3: Translate forecast parameters into mathematical safety stock metrics and automated dynamic Reorder Points (*ROP*).

2.4: Build an evaluation interface/dashboard to visually validate model outputs against actual historical hold patterns.

- **Contribution of the Project**

This project contributes academically by providing an empirical blueprint on how end-to-end data engineering pipelines directly operationalize machine learning models inside the logistics sector—bridging the gap between isolated algorithmic design and real-world execution. Mechanically, the framework helps wholesale distributors lower holding expenditures, prevent waste due to expired drug stocks, and secure consistent, uninterrupted medicine access for downstream clinics and pharmacies.

3. State of the Art Review

The academic landscape surrounding supply chain management increasingly characterizes the pharmaceutical sector as highly volatile due to epidemiological waves and complex regulatory governance. Historically, statistical time-series forecasting frameworks like Autoregressive Integrated Moving Average (ARIMA) and its seasonal variant (SARIMA) formed the baseline for inventory planning. While computationally lean and highly interpretable, literature demonstrates that these linear models perform poorly when dealing with intermittent, sparse order patterns typical of specialized or high-cost medications.

To counter these limitations, recent machine learning literature highlights gradient-boosted decision trees (such as XGBoost) and deep learning long short-term memory (LSTM) networks as superior frameworks for capturing sharp, non-linear demand shifts. Supervised machine learning allows for the integration of multidimensional exogenous variables—such as regional demographics and historical supplier delays—significantly reducing Symmetric Mean Absolute Percentage Error (SMAPE) relative to linear models.

However, a distinct gap persists within current research: the majority of academic papers analyze model tuning inside highly sanitized environments using isolated notebook files. There is a lack of focus on the **end-to-end data engineering workflows** necessary to handle structural data drift, stream ingestion, and the conversion of predictive arrays into actionable inventory logic (such as safety stock variations). This thesis directly bridges

this gap by demonstrating an operationalized system using a continuous 5-year historical ledger.

4. Methods

This study adopts an empirical, data-driven methodology categorized into three major components:

- **Data Acquisition & Engineering:** The primary data source consists of 5 years of historical order records from a wholesale pharmaceutical distributor. The data will undergo systematic preprocessing (imputation of null entries, duplicate elimination, and formatting standardization). Features will be engineered by creating time-lag indicators (e.g., previous 7-day, 30-day volumes) and categorical flags matching core drug families.
- **Algorithmic Modeling Framework:** The forecasting task will be modeled as a supervised regression problem. We will deploy and tune an **XGBoost Regressor** pipeline alongside a **SARIMA** baseline. The optimization objective will prioritize reducing error parameters across varying operational horizons (weekly vs. monthly predictions).
- **Inventory Optimization Logic:** The predicted demand values (D) and calculated lead times (L) will be programmatically fed into safety stock equations to calculate dynamic Reorder Points (ROP) via:

$$ROP = (Average\ Daily\ Demand \times Lead\ Time) + Safety\ Stock$$

This ensures the predictive model translates cleanly into real-world business actions.

5. Proposed Work

5.1 Intended contribution of the work

The primary deliverable is an automated pipeline that accepts historical, multi-product transactional inputs and transforms them into predictive stocking triggers. It integrates content-based grouping, where drug families with low individual frequencies are aggregated into broader categories to overcome data sparsity challenges during model training.

5.2 Benefits of the proposed work

The primary beneficiaries are wholesale pharmaceutical managers, logistical planners, and healthcare institutions. Implementing this system minimizes capital lock-up in sluggish inventories, decreases disposal overheads from expired pharmaceuticals, and reduces the probability of regional drug shortages.

5.3 Procedures and activities of the proposed work

Objective	Activity	Milestone
Data Architecture	Ingest, clean, and map the 5-year historical distribution dataset.	Preprocessed dataset and feature schema finalized.
Model Prototyping	Train, optimize, and cross-validate ARIMA and XGBoost architectures.	Completed model training runs with optimized hyper-parameters.
Inventory Synthesis	Code the dynamic reorder layer using model prediction coefficients.	Inventory engine logic fully validated.
Operational Validation	Evaluate error distributions and display metrics via an analytical layout.	Analytical visualization completed and documented.

5.4 Evaluation of the project

- **Quantitative Model Evaluation:** Predictive accuracy will be rigorously evaluated using root mean squared error ($RMSE$), mean absolute error (MAE), and R^2 scores across training sets.
- **Operational Simulation:** The inventory optimization logic will be back-tested against the historical ledger to measure simulated reductions in overstock volumes and stockout events.
- **Process Review:** Regular development check-ins alongside academic advisors will verify architectural adherence to scalable, best-practice data engineering design principles.

5.5 Resources needed

- **Software Environment:** Python 3.x ecosystem (Pandas, NumPy, Scikit-Learn, XGBoost, Stats models).
- **Database Infrastructure:** Local or cloud-based relational database instances (PostgreSQL / SQLite).
- **Hardware Compute:** Standard workstation with multi-core CPU capabilities for parallelizing hyper-parameter grid searches.

5.6 Access to participants and clients

The project relies on an existing, anonymized 5-year transaction history log, bypassing the need for live human participant recruitment.

5.7 Ethical Aspects

The transaction ledger is strictly restricted to business-to-business (B2B) operational metrics (SKU counts, product identifiers, anonymized client location codes, and transaction timestamps). It contains **no identifiable patient information, personal medical prescriptions, or sensitive individual records**, ensuring complete alignment with standard university data safety governance.

6 Work plan and schedule

The plan is created on **Smartsheet** a cloud project management tool, and an **excel export** of the **WBS** and **Gantt chart** are provided to illustrate the timeline and milestones.

Phase/Objective	Activities	Milestone
Phase1: Initiation		
	Project scope definition and architecture mapping	Proposal
	Initial database structural exploration	
Phase2: Literature Review		
	Research time-series and demand forecasting papers	Proposal and thesis
	Synthesize algorithmic comparison summaries	
Phase3: Requirements Analysis & Design		
	Dataset collection phase	Feature Warehouse Built
	Data pipeline schema mapping and ETL design	
	Data cleansing and missing vector management	
Phase4: Development		
	Feature engineering & lag creation	Fully Trained Forecasting Engine
	Baseline (SARIMA) vs ML (XGBoost) modeling	
Phase5: Documentation & Graded Deliverables		
	proposal draft	Thesis
	final proposal	
	formal project proposal	
	Ch1 introduction partial and ch2 literature review	

	submission, approval, and feedback	
	final report	
	viva voice plus poster	

7 Conclusions

This proposal outlines a robust, dual-discipline project that directly satisfies the advanced competencies of an MSc in Data Engineering and AI. By converting 5 years of historical pharmaceutical distribution records into a live, predictive optimization mechanism, the research addresses real-world industrial inefficiencies. The systematic structure detailed within this document ensures clear execution and straightforward tracking as the project advances toward its final delivery.

8 Bibliography

- Ahmed, A. A. A., 2022. Integrated artificial intelligence effect on crisis management and lean production: structural equation modelling frame work. *Springer Nature*, p. 220–227.
- Devlin, J., 2019. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. s.l., Association for Computational Linguistics.
- Mnih, V., 2015. Human-level control through deep reinforcement learning. *nature*, p. 529–533.
- Petropoulos, F., 2022. Forecasting: Theory and Practice. *nature*, pp. 705-871.
- Radford, A., 2019. *Language Models are Unsupervised Multitask Learners*, s.l.: OpenAI.
- Rathipriya, R., 2022. Deep Learning-Based Demand Forecasting Architecture for Supply Chain Management. *Computers and Electrical Engineering*.
- Silver, D., 2016. Mastering the game of Go with deep neural networks and tree search. *Nature*, p. pages484–489.
- Yu, L., 2021. Reinforcement learning approach for resource allocation in humanitarian logistics. *Expert Systems with Applications*.

9 Appendixes

- WBS

Health	Primary	Start Date	End Date	% Completion
Project Information				
Yellow	Project Plan	25-Jun-26	01-Jul-26	60%
Green	Phase 1: Project Initiation & Data Exploration	05-Jul-26	20-Jul-26	60%
Green	Discovery on pharmaceutical supply chain solutions & research	06-Jul-26	17-Jul-26	100%
Green	Explore data sources and database constraints (5-year invoice ledger)	08-Jul-26	24-Jul-26	80%
Green	Scope definition and final parameter mapping	09-Jul-26	25-Jul-26	50%
Green	Phase:2 Literature review	20-Jun-26	09-Jul-26	100%
Green	Research time-series & GBDT papers on Google Scholar / IEEE	21-Jun-26	09-Jul-26	100%
Green	Research supply chain operationalization architectures	22-Jun-26	09-Jul-26	100%
Blue	Phase:3 Requirements analysis & Design	25-Jul-26	20-Aug-26	0%
Blue	Database initialization and ingestion schema configuration (SQL)	25-Jul-26	05-Aug-26	0%
Blue	Prototype ETL logic—prioritize SKU-level description classifications	30-Jul-26	10-Aug-26	0%
Blue	Prototype temporal flag mapping (Date vectors and QTY scaling parameters)	06-Aug-26	16-Aug-26	0%
Blue	Prototype baseline exploratory tracking charts	15-Aug-26	20-Aug-26	0%
Blue	Phase:4 Design	25-Aug-26	05-Sep-26	0%

Project Proposal: The All-Rounder

Blue	Design system context and automated data engineering pipeline diagrams	25-Aug-26	30-Aug-26	0%
Blue	Model execution pipeline blueprint architecture	31-Aug-26	05-Sep-26	0%
Blue	Phase:5 Development	06-Sep-26	27-Sep-26	0%
Blue	Complete pipeline automation script assembly	06-Sep-26	11-Sep-26	0%
Blue	Train and evaluate baseline time-series frameworks (SARIMA)	12-Sep-26	18-Sep-26	0%
Blue	Program, evaluate, and cross-validate the supervised XGBoost Regressor	19-Sep-26	25-Sep-26	0%
Blue	Interface creation (Power BI dashboard layer to visualize forecast variances)	25-Sep-26	27-Sep-26	0%
Green	Phase:6 Documentation & Graded Deliverables	25-Jun-26	01-Oct-26	10%
Green	Initial proposal draft compilation	25-Jun-26	01-Jul-26	100%
Green	Final project proposal generation and layout assembly	01-Jul-26	10-Jul-26	100%
Green	Chapter 1 (Introduction) and Chapter 2 (Literature Review) submission	20-Jun-26	09-Jul-26	100%
Blue	Academic mid-point review evaluation and milestone review check	10-Aug-26	10-Aug-26	0%
Blue	Final comprehensive report compilation and technical write-up	28-Sep-26	28-Sep-26	0%
Blue	Viva voce presentation practice and final poster delivery	01-Oct-26	01-Oct-26	0%